Mathematical definitions and conventions

$$Y_{S_1} = \frac{\alpha}{2}, \qquad Y_F = \frac{\alpha+1}{2}, \qquad Y_{S_2} = \frac{\alpha+2}{2},$$

$$j(d) = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12},$$

$$D_\mu \Phi = \left(\partial_\mu + i g_1 Y_\Phi B_\mu + i g_2 T^a_\Phi W^a_\mu\right) \Phi,$$

$$\widetilde{H} = i\sigma^2 H^*, \qquad \beta_X = \mu \frac{dX}{d\mu}, \qquad \beta_X = \frac{\beta_X^{(1)}}{16\pi^2} + \cdots,$$

$$T_\nu^{(1)} = \mathrm{Tr}(y_1^\dagger y_1), \qquad T_\nu^{(2)} = \mathrm{Tr}(y_2^\dagger y_2),$$

$$T_\nu = T_\nu^{(1)} + T_\nu^{(2)}, \qquad T = \mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d).$$

$$\begin{aligned}
\mathcal{L}_{Y,\mathrm{SM}} &\supset -\bar{Q} Y_u \widetilde{H} u - \bar{Q} Y_d H d - \bar{L} Y_e H e + \mathrm{h.c.}, \\
V &\supset m_{S_1}^2 S_1^\dagger S_1 + m_{S_2}^2 S_2^\dagger S_2 + \lambda_1 (H^\dagger H)^2 \\
&\quad + \lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1) + \lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2) \\
&\quad + \lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2), \\
V_{\mathrm{adj}} &\supset \lambda_{HS_1}^{(A)} (H^\dagger T^a H)(S_1^\dagger T^a S_1) + \lambda_{HS_2}^{(A)} (H^\dagger T^a H)(S_2^\dagger T^a S_2) \\
&\quad + \lambda_{12}^{(A)} (S_1^\dagger T^a S_1)(S_2^\dagger T^a S_2), \\
V_\times &\supset \lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1).
\end{aligned}$$

T3 Weinberg-operator coefficient comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$

$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5 C_{LLS_1S_2}$

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-A-m4	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-A-m2	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-A-p0	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-B-m3	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-B-m1	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-B-p1	$A_5^{ij}=$	$C_5^{ij}=$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$\frac{2}{3}+\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-C-m3	$A_5^{ij}=$	$C_5^{ij}=$
	$-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$-\frac{2}{3}+\left(\frac{\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-C-m1	$A_5^{ij}=$	$C_5^{ij}=$
	$-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$	$-\frac{2}{3}+\left(\frac{\hbar\overline{y_{1,i,r}\sqrt{3}\hbar\overline{y_{2,j,r}}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-C-p1	$A_5^{ij} =$	$C_5^{ij} =$
	$-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-D-m2	$A_5^{ij} =$	$C_5^{ij} =$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-D-p0	$A_5^{ij} =$	$C_5^{ij} =$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-D-p2	$A_5^{ij} =$	$C_5^{ij} =$
	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-E-m4	$A_5^{ij} =$	$C_5^{ij} =$
	$-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-E-m2	$A_5^{ij} =$	$C_5^{ij} =$
	$-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-E-p0	$A_5^{ij} =$	$C_5^{ij} =$
	$-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$
T3-E-p2	$A_5^{ij} =$	$C_5^{ij} =$
	$-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$	$\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$